

Preliminary results for Humboldt marten WGS project

March 31, 2026

We have resequenced whole genomes from 20 Humboldt martens (*Martes caurina humboldtensis*) from 4 of their populations: Oregon North Dunes (“OR Dunes N”; n=7), Oregon South Dunes (“OR Dunes S”; n=5), Southern Oregon (“OR South”; n=4), and Northern California (“CA North”; n=4). We have also resequenced whole genomes from 10 montane Pacific martens (*Martes caurina caurina*) in the Oregon Central Cascades (“OR Cascades”) (Figure 1). In addition to these samples, we are taking advantage of publicly available whole genome sequence (WGS) data from 2 montane Pacific martens from Washington and Colorado, as well as 2 American martens (*Martes americana*) from Alaska (Colella et al 2021).

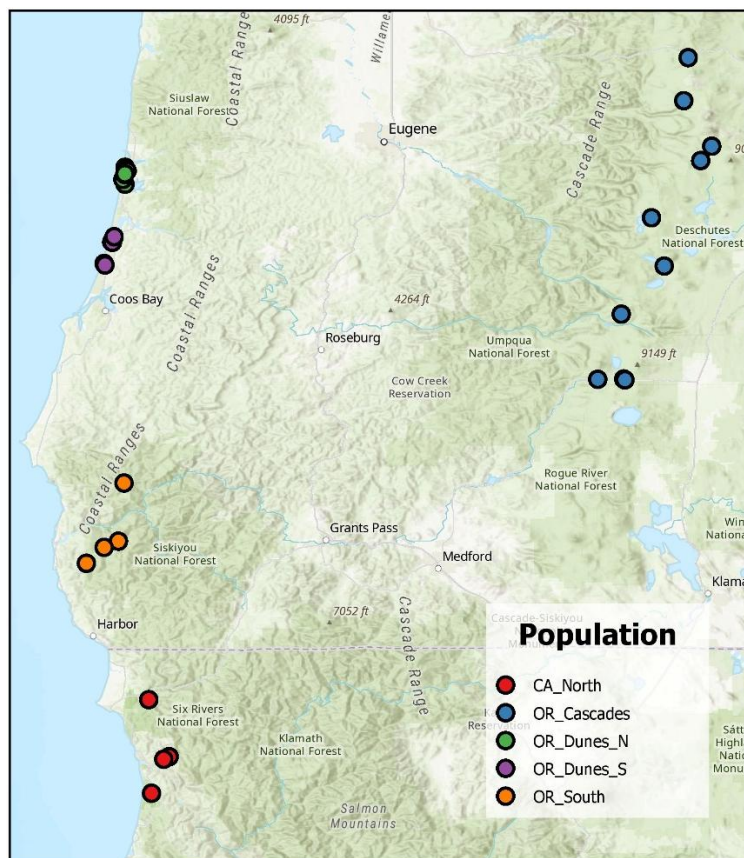


Figure 1. Map of Humboldt marten and montane Pacific marten samples from Oregon and California.

Here, I present preliminary results from my first objective, which is to describe the genetic diversity and connectivity of Humboldt marten populations. The results are preliminary because I have mapped our WGS data to the yellow-throated marten assembly (*Martes flavigula*; GCA_029410595.1) as I wait for the completion of the Humboldt marten assembly. Though *M. flavigula* is fairly divergent from Humboldt marten, we chose the assembly for its high quality. We expect that some results may change when our data is mapped to the Humboldt assembly but hope that general patterns hold.

We chose to reduce our mapping quality threshold from 30 to 20 in order to reduce “reference bias,” a phenomenon where use of a diverged reference results in the under-estimation of genetic diversity. Using a mapping quality threshold of 20, we had 89-93% of sequencing reads successfully mapping to the *M. flavigula* assembly, resulting in 16.5-22.0X coverage (average 19X). Because the additional samples from Colella et al 2021 were considerably higher coverage, they were down-sampled to 19X to be more comparable to our samples. One sample from CA North was removed from analysis due to suspected contamination (low mapping success in addition to abnormally high heterozygosity). Individual VCF files were filtered by removing: SNPs within 5 bp of an indel, indels, genotypes with genotype quality < 30, sites with < 6 reads, heterozygous sites with allelic imbalance (< 20% support for minor allele), sites with >2X the mean pooled read depth, sites with low mappability (mappability mask created by genmap), and sex chromosomes. Individual VCFs were merged and only biallelic SNPs with a minor allele count of at least 1 were retained. Additional filtering removed sites where >80% of samples were heterozygous, sites with QUAL phred score < 30, and sites where >20% samples had missing genotypes. After filtering, we have 4,037,356 high quality SNPs.

Population Structure

For PCA and Admixture plots, we additionally thinned our data by removing SNPs in linkage disequilibrium.

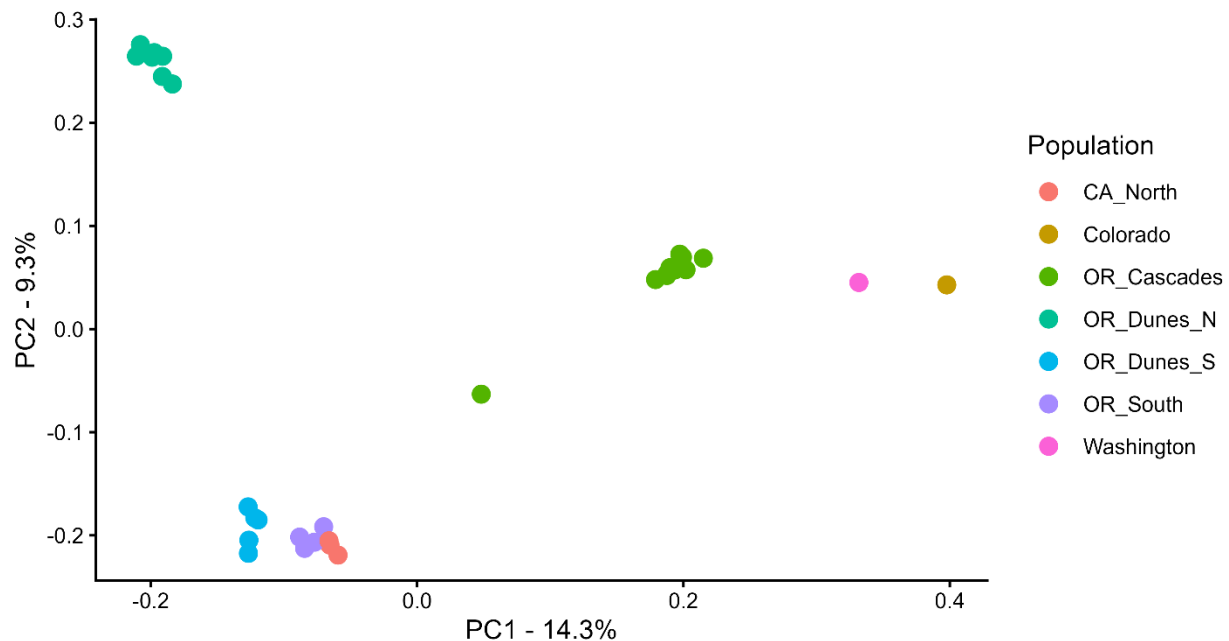


Figure 2. PCA plot of Humboldt martens and montane Pacific martens. Humboldt martens are separated from montane Pacific martens along the first principal component. One Oregon Cascades sample appears intermediate between the Oregon Cascades cluster and the Southern Oregon / Northern California cluster in principal component space. Oregon Dunes North clusters separately from other populations along the second principal component.

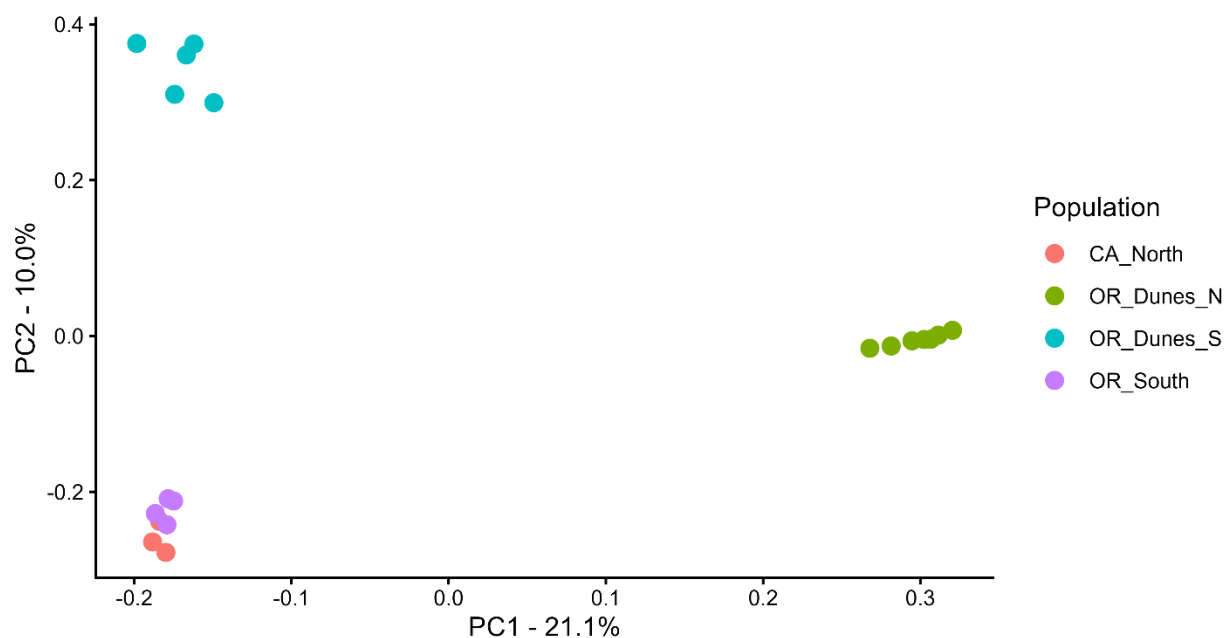
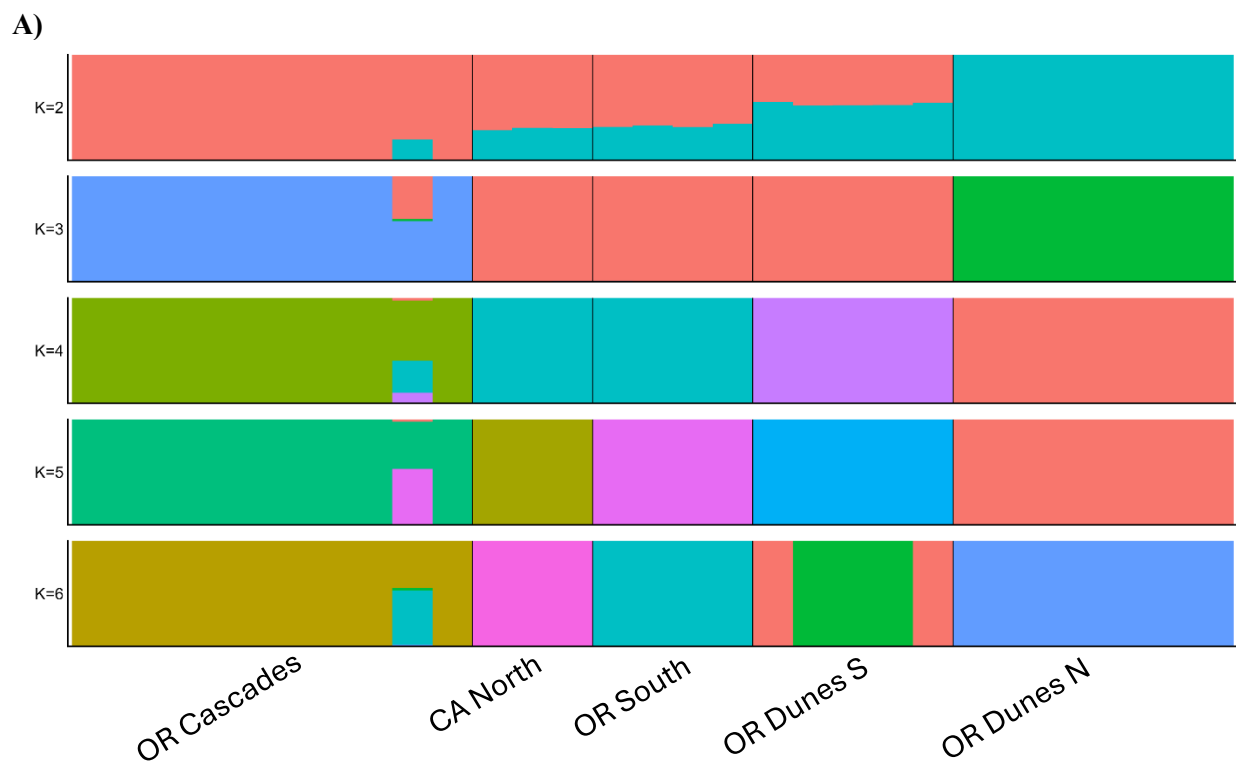


Figure 3. PCA plot of just Humboldt martens. Oregon Dunes North clusters separately from other populations along the first principal component, and Oregon Dunes South now clusters separately from the Southern Oregon / Northern California cluster along the second principal component.



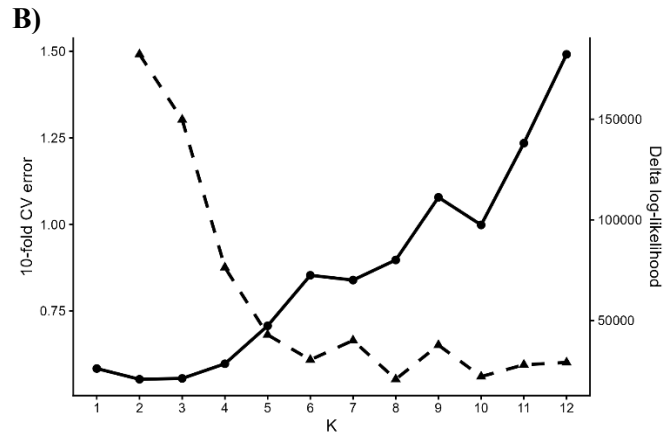
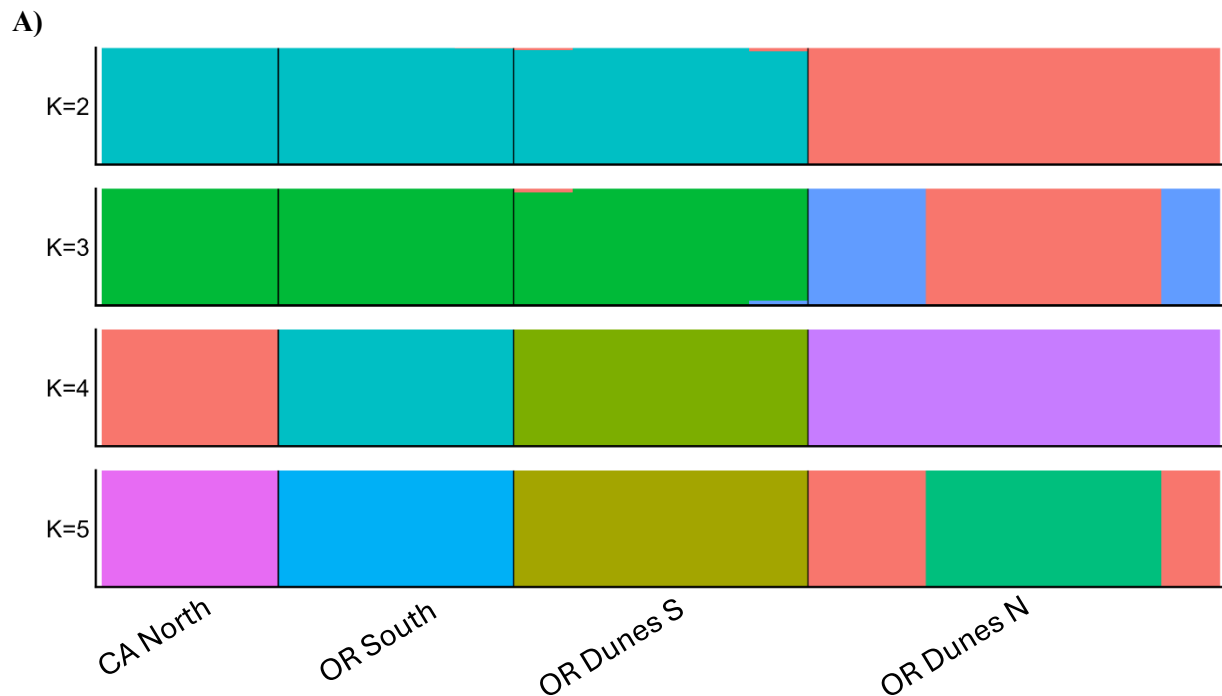


Figure 4. A) Admixture plot of Humbolt martens and montane Pacific martens from the Oregon Cascades. Oregon Dunes North immediately separates out as its own genetic cluster, likely due to high relatedness among those individuals. K=5 cleanly separates the five populations into their own genetic clusters, except for the individual from the Oregon Cascades with intermediate ancestry. **B)** 10-fold cross validation error and delta log-likelihood showing K=2 as the “best” model.



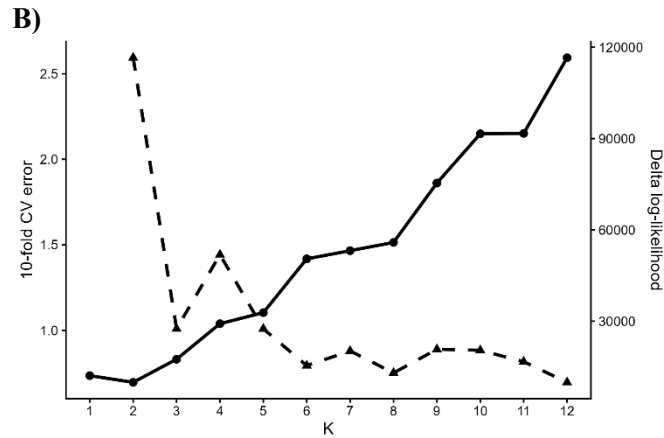


Figure 5. A) Admixture plot of just Humboldt. Again, Oregon Dunes North immediately separates out as its own genetic cluster, likely due to high relatedness among those individuals. K=4 cleanly separates the four populations into their own genetic clusters. **B)** 10-fold cross validation error and delta log-likelihood showing K=2 as the “best” model.

pop	pop sample	OR_Dunes_S 3571	OR_Dunes_S 16	OR_Dunes_S 3574	OR_Dunes_S 3577	OR_Dunes_S 3579	OR_Dunes_N 3573	OR_Dunes_N 3568	OR_Dunes_N 3569	OR_Dunes_N 3572	OR_Dunes_N 3575	OR_Dunes_N 3576
OR_Dunes_S	3571											
OR_Dunes_S	16	0.067										
OR_Dunes_S	3574	0.031	0.069									
OR_Dunes_S	3577	0.006	0.087	0.141								
OR_Dunes_S	3579	-0.044	0.012	0.038	0.094							
OR_Dunes_N	3573	-0.430	-0.324	-0.408	-0.504	-0.561						
OR_Dunes_N	3568	-0.466	-0.340	-0.418	-0.528	-0.596	0.182					
OR_Dunes_N	3569	-0.365	-0.260	-0.336	-0.431	-0.469	0.276	0.104				
OR_Dunes_N	3572	-0.385	-0.273	-0.350	-0.453	-0.505	0.048	0.121	0.021			
OR_Dunes_N	3575	-0.405	-0.294	-0.368	-0.483	-0.538	0.030	0.052	0.027	0.241		
OR_Dunes_N	3576	-0.393	-0.282	-0.379	-0.470	-0.529	0.129	0.127	0.085	0.188	0.297	
OR_Dunes_N	3578	-0.581	-0.432	-0.531	-0.660	-0.742	-0.104	-0.129	-0.030	0.035	0.274	0.151

Figure 6. KING kinship coefficients for sample pairs in Oregon Dunes North and Oregon Dunes South. First-degree relatives (parent-offspring / full sib) have a KING kinship coefficient ~0.25 and second-degree relatives (half sib) have a KING kinship coefficient ~0.125. Oregon Dunes South appears to have one second-degree relative pair, while Oregon Dunes North has several first and second-degree relative pairings. Inbreeding is likely inflating kinship coefficients in this population. Only Oregon Dunes North and Oregon Dunes South had first or second-degree relatives.

	M_americana	COWA	OR_Cascades	CA_North	OR_South	OR_Dunes_S
M_americana						
COWA	0.737					
OR_Cascades	0.815	0.167				
CA_North	0.813	0.249	0.156			
OR_South	0.824	0.270	0.160	0.053		
OR_Dunes_S	0.850	0.359	0.211	0.189	0.154	
OR_Dunes_N	0.879	0.460	0.275	0.349	0.317	0.352

Figure 7. Pairwise F_{ST} (Weir and Cockerham weighted) between populations. Pacific marten samples from Washington and Colorado (Colella et al 2021) were grouped together here as “COWA.” Notably, Oregon Dunes South and Oregon Dunes North have relatively high F_{ST} and Northern California and Southern Oregon have relatively low F_{ST} .

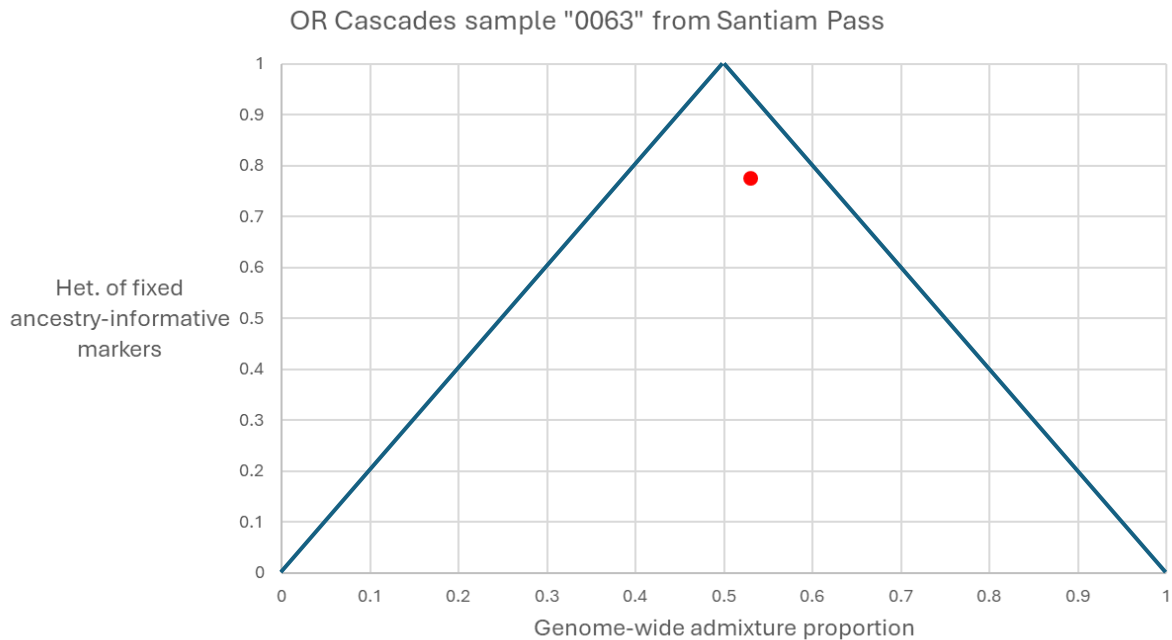


Figure 8. Triangle plot of interclass heterozygosity vs. hybrid index for putatively admixed Oregon Cascades individual to assess hybrid status. Interclass heterozygosity is defined as the heterozygosity of fixed ancestry-informative markers (AIMs) for Humboldt martens and montane Pacific martens. Hybrid index is defined as the genome-wide admixture proportion (taken from Admixture analysis), where 0 indicates no Oregon Cascades ancestry and 1 indicates 100% Oregon Cascades ancestry. An F1 hybrid is expected to plot at the top of the triangle with 50% Oregon Cascades ancestry and 100% heterozygosity of fixed AIMs. F2 hybrids are expected to plot around the middle of the triangle, and back crosses to either parent species are expected to plot along the sides of the triangle. The putatively admixed individual from the Oregon Cascades (in red) doesn’t cleanly fall into any of these categories, though we can likely rule out a back cross. With limited samples representing our parent species, we can’t know what SNPs are truly fixed between the species with high confidence; thus, some of the SNPs included in this analysis may not truly be fixed. Acknowledging this fact, it is possible that this individual is an F1 hybrid.

Genomic diversity

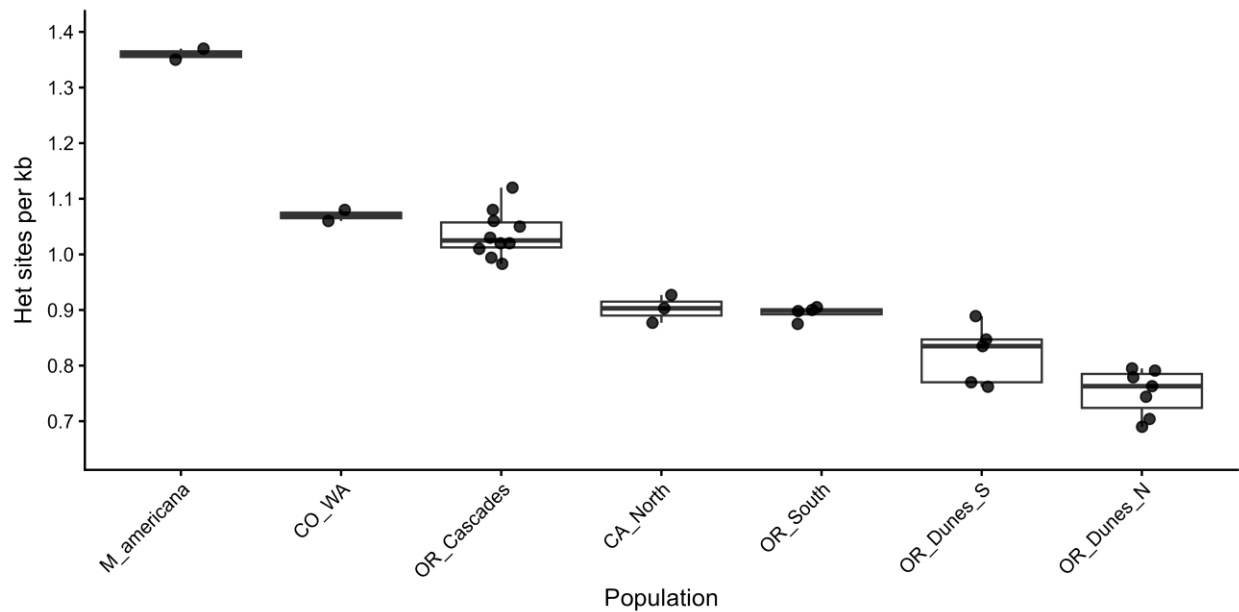


Figure 9. Genome-wide heterozygosity defined as heterozygous sites per kilobase, estimated from mlRho. Humboldt martens overall have lower heterozygosity than the montane Pacific martens and American martens, and the Oregon Dunes North population specifically has the lowest heterozygosity.

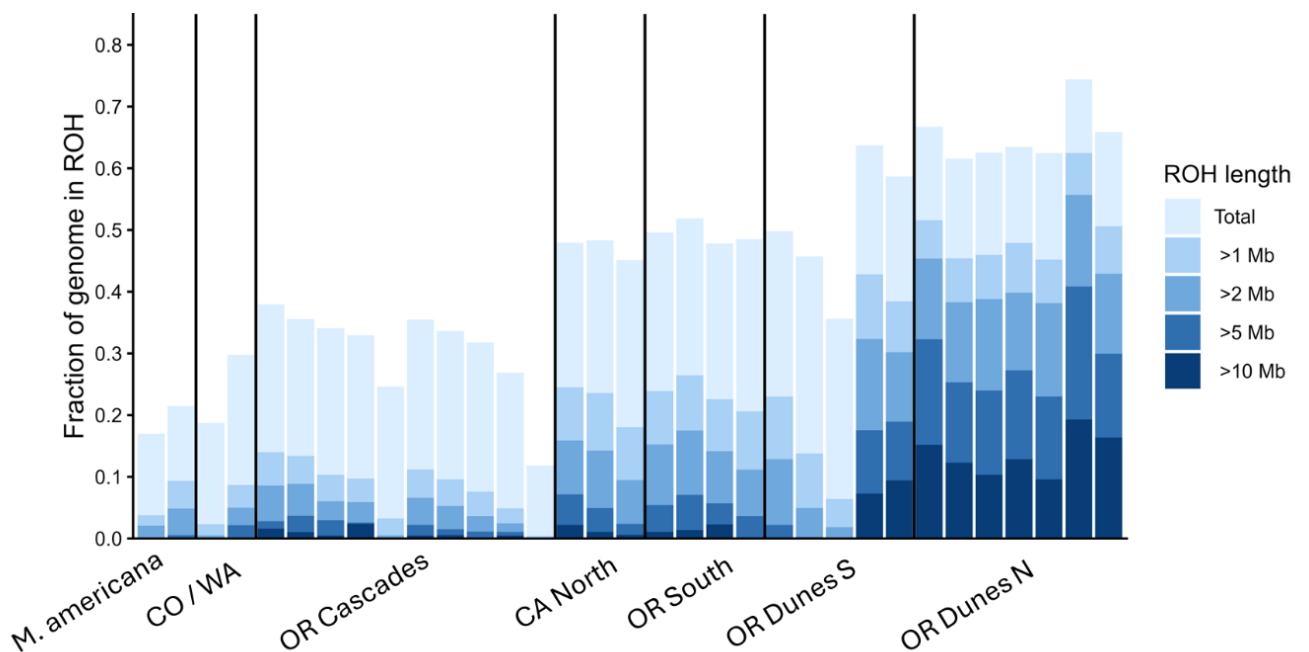


Figure 10. Inbreeding, measured by fraction of the genome occurring within a run of homozygosity (fROH) estimated by plink. fROH is parsed into various length thresholds, with longer ROH indicating inbreeding between close relatives. Oregon Dunes North has the highest fROH overall, and notably it's the only population where every individual has some ROH greater than 10 Mb (considered to be very long ROH).

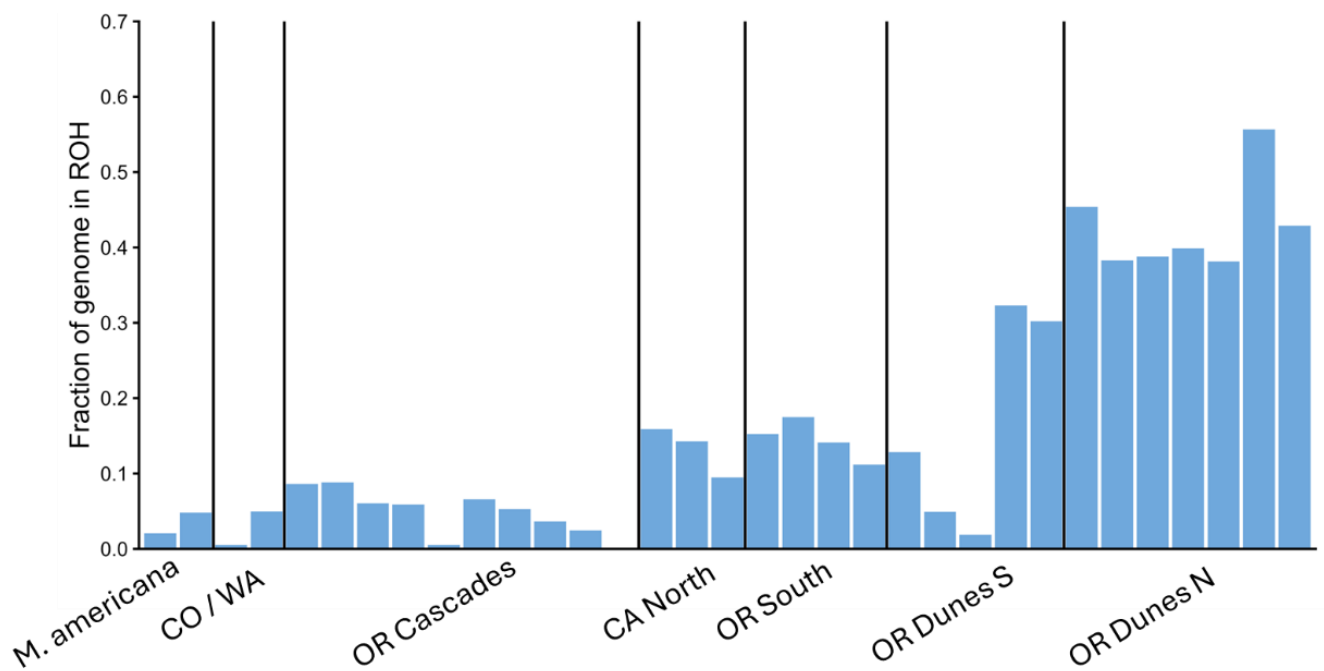


Figure 11. Inbreeding between close relatives, measured by fraction of the genome occurring within a long run of homozygosity (fROH > 2Mb) estimated by plink. This is the same data from Figure 10, but showing just the >2Mb category. In mammals, assuming a standard recombination rate of 1 cM/Mb, ROH 2Mb or longer arises from parents sharing a common ancestor less than 25 generations ago. Because recessive deleterious mutations segregate in mammals for 5-25 generations on average, ROH 2Mb or longer may contain these recessive mutations in the homozygous state, where they may impact fitness (Kyriazis et al. 2025).

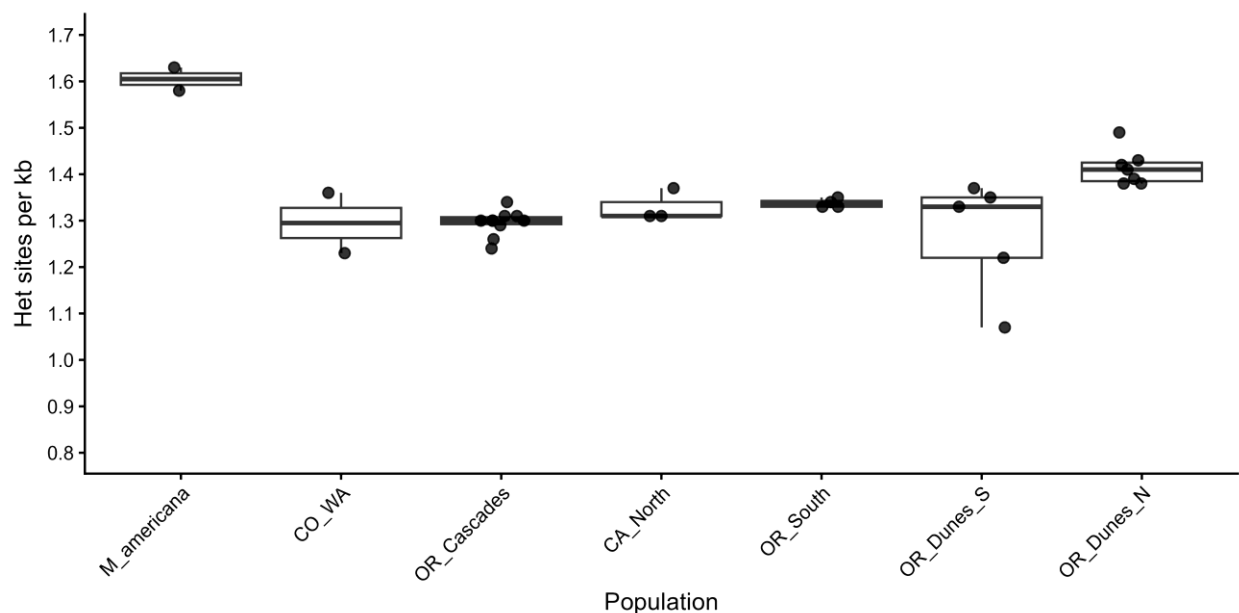


Figure 12. Non-ROH heterozygosity (heterozygosity in non-ROH regions of the genome), defined as heterozygous sites per kilobase, estimated from mlRho. Non-ROH heterozygosity allows us to disentangle the impact of inbreeding and drift on reduced genomic diversity. While there were clear differences between Humboldt marten populations in genome-wide heterozygosity, the populations seem to have similar non-ROH heterozygosity. This suggests that inbreeding is playing a major role in reducing genomic diversity in Humboldt martens.

Inbreeding Depression Risk

Inbreeding depression risk can be assessed through fraction of the genome in long ROH and non-ROH heterozygosity (Kyriazis et al 2025). Because recessive deleterious mutations typically “hide” in the heterozygous state, where they do not impact fitness, non-ROH heterozygosity is indicative of deleterious variation that may become expressed through inbreeding. The ID_{risk} statistic is calculated by taking $fROH > 2Mb$ and multiplying by non-ROH heterozygosity (heterozygous sites per kb). Values 0.5 or greater indicate extreme risk of inbreeding depression.

Table 1. ID_{risk}

population	fROH_2Mb	nonROH_het	ID_risk
M_americana	0.0344	1.605	0.06
CO_WA	0.0273	1.295	0.04
OR_Cascades	0.0479	1.295	0.06
CA_North	0.1320	1.330	0.18
OR_South	0.1451	1.338	0.19
OR_Dunes_S	0.1643	1.268	0.21
OR_Dunes_N	0.4271	1.414	0.60

Following this statistic, Oregon Dunes North is at extreme risk of experiencing inbreeding depression, while Oregon Dunes South, Southern Oregon, and Northern California are at moderate risk. Oregon Cascades has considerably lower ID_{risk} than the Humboldt marten populations, and is just barely at moderate risk for experiencing inbreeding depression.

Conclusions

Humboldt martens clearly separate from montane Pacific martens in principal component space and in Admixture analysis, with the exception of a putatively admixed individual from the Oregon Cascades. Within Humboldt martens, there is clear structure between the populations, although Northern California and Southern Oregon consistently cluster together. No other individuals show evidence of intermediate ancestry between the populations, suggesting a lack of connectivity. High relatedness and inbreeding in Oregon Dunes North likely drives the extreme differentiation of that population from other Humboldt marten populations. In the future, I plan to reduce the Oregon Dunes populations to a subset of non-related individuals and re-run Admixture and PCA. Humboldt martens have reduced heterozygosity relative to montane Pacific martens, which can be explained by higher inbreeding in their populations. Oregon Dunes North has especially high inbreeding as indicated by long runs of homozygosity, and is at extreme risk of experiencing inbreeding depression.